Dietary Restriction, Growth Factors and Aging: from yeast to humans

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Dietary restriction (DR) and reduced growth factor signaling both elevate resistance to oxidative stress, reduce macromolecular damage, and increase lifespan in model organisms. In rodents, both DR and decreased growth factor signaling reduce the incidence of tumors and slow down cognitive decline and aging. DR reduces cancer and cardiovascular disease and mortality in monkeys, and reduces metabolic traits associated with diabetes, cardiovascular disease and cancer in humans. Neoplasias and diabetes are also rare in humans with loss of function mutations in the growth hormone receptor. DR and reduced growth factor signaling may thus slow aging by similar, evolutionarily conserved, mechanisms. We review these conserved anti-aging pathways in model organisms, discuss their link to disease prevention in mammals, and consider the negative side effects that might hinder interventions intended to extend healthy lifespan in humans.